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1.(24 pts)

Character	Hollerith	EBCDIC binary	EBCDIC hex	ASCII binary	ASCII hex	ASCII Decimal
C	12-3 ✓	1100 0011 ✓	C3 ✓	0100 0011	43 ✓	67 ✓
K	11-2 ✓	1101 1011 0010	B2 ✓	0100 1011	4B ✓	51 75
T	0-3 ✓	1110 0000 0011	D3 E3	0101 0011 ✓	53 64	83 84
7	7 ✓	1111 0111 ✓	F7 ✓	0011 0111	37	55 55

A=41
 B=42
 C=43
 D=44
 E=45
 F=46
 G=47
 H=48
 I=49
 J=4A
 K=4B
 Q=50
 T=53

2.(16 pts) Perform the below conversions from decimal to the appropriate base

Decimal	8-bit binary	hex	Base 5	Base 7
17	00010001	11	32	23
39	0010 0111 ✓	27 ✓	74 X 124	54 ✓
69	0100 0101 ✓	45 ✓	234 ✓	126 ✓
102	0110 0110 ✓	66 ✓	402 ✓	204 ✓
127	0111 1111 ✓	7F ✓	502 ✓	241 2

3

3.(16) Perform the below additions in the appropriate bases. You may assume that you can include the "carry-out".

$$\begin{array}{r}
 10011011_2 \\
 +01101111_2 \\
 \hline
 100001010
 \end{array}$$

$$\begin{array}{r}
 30442_5 \\
 +24334_5 \\
 \hline
 110331
 \end{array}$$

$$\begin{array}{r}
 3564_7 \\
 +4536_7 \\
 \hline
 10433
 \end{array}$$

$$\begin{array}{r}
 2C3AB_{16} \\
 +FE28D_{16} \\
 \hline
 1A658
 \end{array}$$

$$\begin{array}{r}
 10011011_2 \\
 -01101111_2 \\
 \hline
 0010100
 \end{array}$$

$$\begin{array}{r}
 30442_5 \\
 -24334_5 \\
 \hline
 1103
 \end{array}$$

$$\begin{array}{r}
 3564_7 \\
 -2536_7 \\
 \hline
 1025
 \end{array}$$

$$\begin{array}{r}
 DC3AB_{16} \\
 -AE28D_{16} \\
 \hline
 2E1E
 \end{array}$$